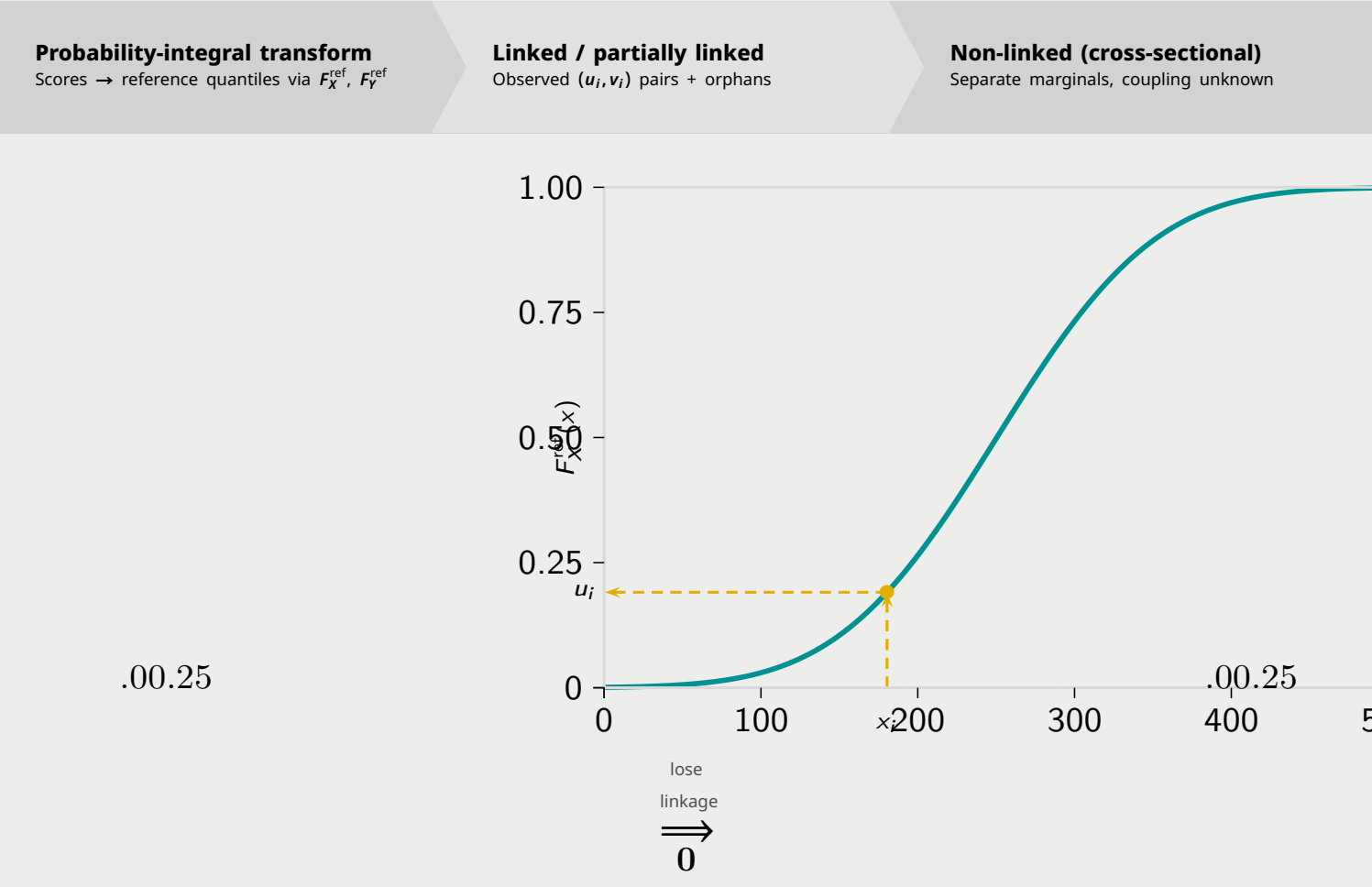


From Scores to Pseudo-Observations: The Marginal Transformation

Stripping Metric Structure via the Probability-Integral Transform

Linked, partially linked, and non-linked observations in copula-based longitudinal inference without longitudinal data



The probability-integral transform. Suppose we observe prior scores $X = (x_1, \dots, x_{n_X})$ and current scores $Y = (y_1, \dots, y_{n_Y})$ for a subgroup S (e.g., a TIMSS country or a NAEP state). The scores live on assessment-specific metric scales whose units carry no intrinsic cross-time meaning. To place both margins on a common, metric-free footing we apply fixed *reference* cumulative distribution functions (CDFs), computed over the full assessment population, to obtain *pseudo-observations* on $[0, 1]$:

$$u_i = F_X^{\text{ref}}(x_i), \quad v_j = F_Y^{\text{ref}}(y_j), \tag{M.1}$$

where F_X^{ref} and F_Y^{ref} are the prior-grade and current-grade population CDFs, respectively. This is the classical *probability-integral transform* (PIT): if F_X^{ref} is the true marginal CDF of X , then $U = F_X^{\text{ref}}(X)$ is uniformly distributed on $[0, 1]$.

Population versus subgroup. The reference CDFs are defined over the *full population*, not over S . Consequently, the marginal distribution of U within subgroup S is generally *not* uniform—it reflects S ’s position relative to the population. A subgroup that is below-average on prior achievement will have a density of U concentrated in the lower quantiles. The population-level Uniform(0, 1) serves as the baseline against which subgroup departures are measured.

Linkage as observed coupling. When prior and current scores come from the *same students* measured longitudinally, we observe pairs (x_i, y_i) —and hence pairs (u_i, v_i) —that carry information about how achievement at two time points co-varies within individuals. Formally, linked data are draws from a joint law,

$$(U_i, V_i) \stackrel{\text{i.i.d.}}{\sim} C(F_X^{\text{ref}}(x), F_Y^{\text{ref}}(y)), \tag{M.2}$$

where C is the copula encoding within-subject dependence (Sklar’s theorem). The spaghetti plot on the left makes this coupling visible: each horizontal line connects a student’s prior quantile u_i to their current quantile v_i .

Partial linkage. In practice, matching rates are rarely 100%. Some students appear in the prior sample but not the current one (u_i with no corresponding v_i), and vice versa. These *orphan observations*—shown as $\times$ marks in the linked panel—still contribute to the marginal distributions of U and V but carry no direct coupling information.

Non-linkage. Cross-sectional assessment programs (NAEP, TIMSS, PIRLS) draw independent samples at each occasion. The analyst observes only the marginal distributions,

$$X_1, \dots, X_{n_X} \stackrel{\text{i.i.d.}}{\sim} F_X, \quad Y_1, \dots, Y_{n_Y} \stackrel{\text{i.i.d.}}{\sim} F_Y, \tag{M.3}$$

with no within-subject pairing observed. After the PIT, the two dot-strips on the right illustrate this situation: we see where U and V concentrate individually, but the coupling—the copula C that would connect them—is entirely missing and must be supplied by an external model or assumption.

What the marginals preserve and what they discard. The transform $x \mapsto u = F_X^{\text{ref}}(x)$ is monotone and (if F_X^{ref} is continuous) invertible, so no ordinal information is lost.¹ What is discarded is the *metric structure*: the difference $x_2 - x_1$ depends on the scale, but the rank ordering $u_1 < u_2$ is scale-invariant. This is precisely the property exploited by Sklar’s theorem, which factors any joint distribution $H(x, y)$ as

$$H(x, y) = C(F_X(x), F_Y(y)), \tag{M.4}$$

separating the marginal behavior (F_X, F_Y) from the dependence structure (C). By working on the pseudo-observation scale, subsequent inference depends only on the copula—not on the original measurement units.

¹ The transform $x \mapsto u = F_X^{\text{ref}}(x)$ is invertible when F_X^{ref} is continuous (strictly increasing on its support). In discrete or mixed-continuous settings, the CDF is not strictly monotone and the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$ is used. This distinction matters when working with score scales that have mass points (e.g., integer-valued test scores). In such cases, randomized ranks or mid-rank adjustments are sometimes applied to ensure a continuous marginal on $[0, 1]$; see Genest & Nešlehová (2007) and Rüschendorf (2009) for details on pseudo-observation construction under ties.

² The reference CDFs F_X^{ref} and F_Y^{ref} are computed over the *full assessment population*—the complete set of examinees across all subgroups in a given grade and year. They are *not* subgroup-specific. This ensures that all subgroups are mapped to the *same* reference scale, so that a quantile of $u = 0.30$ means “at the 30th percentile of the population” regardless of which subgroup the student belongs to. The subgroup’s marginal density of U then reflects its distributional position within the population.

³ The passage from linked to non-linked data erases the copula C from the observable record. In the linked case, the analyst observes draws from $C(F_X^{\text{ref}}(x), F_Y^{\text{ref}}(y))$ and can estimate C empirically. In the non-linked case, C must be supplied externally—typically from canonical copula tables calibrated on longitudinal datasets in similar content areas and year-spans (see the Overview infographic, Panel B1).

⁴ Partial linkage (matching rates below 100%) occupies an intermediate position: observed pairs contribute direct information about C , while orphan observations contribute only to the marginals. The precision of copula estimation therefore degrades smoothly as the matching rate decreases, reaching the fully non-linked limit when no pairs are observed. The non-linkage premium calculations in the companion appendix quantify this degradation.